

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Shipped **full observability stacks** (OpenTelemetry traces, Prometheus metrics, Grafana dashboards) in production; provisioned cloud infrastructure with **Terraform**; automated deployments on **AWS and GCP**. Built event-driven distributed systems with Kafka and Docker across multiple production environments. Strong Python and TypeScript for tooling automation.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Provisioned cloud infrastructure with **Terraform**; shipped 10+ backend features using TypeScript, Node.js, Express, Prisma ORM, **PostgreSQL**, **Redis**, Auth0, **Docker**, and Sentry across production deployments with CI/CD pipelines.
- Built real-time event pipelines, Temporal.io durable workflow orchestration, and multi-model AI agent tooling (Claude, GPT-4o, Gemini) via MCP; document processing pipelines (PDF gen, email parsing, S3).

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Automated deployments on **AWS and GCP**; integrated **Grafana** monitoring dashboards and Sentry observability across production services; supported model validation and red-teaming for reliability.
- Built AI-driven claim automation (OCR, NLP, ML) improving accuracy 25%.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop distributed pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Full **observability stack** in production: **OpenTelemetry traces + Prometheus metrics** on all hot paths, Grafana dashboards. MySQL outbox pattern, MongoDB TTL projections, **Redis sorted sets** for sub-ms reads, Server-Sent Events for live updates. Load tested with k6; **GitHub Actions CI/CD**.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, OpenTelemetry, Prometheus, Grafana, Docker, GitHub Actions

VectorLift -- Semantic Search & Ranking Engine 2026

Production search on MS MARCO: BM25 (Elasticsearch), dense bi-encoder retrieval, cross-encoder re-ranker. **Kafka-driven indexing pipeline** for event-based corpus ingestion; **per-stage latency tracking** via FastAPI; 6 pipeline configs evaluated with NDCG@10, MRR@10.

Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Streamlit, Docker, Prometheus

Demand Paging -- Xinu OS 2024

Kernel-level virtual memory in C for Xinu OS: 8-store paging, Second-Chance page replacement, inverted page tables, custom fault handler. Added 7+ kernel syscalls; reduced page faults **25%**, memory performance up **15%** -- demonstrates systems-level reliability thinking.

C, Xinu OS, Virtual Memory, Kernel Systems, Page Replacement Algorithms

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

Operating Systems, Algorithms, AI, Data Science, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | DS & Algorithms, OS, Big Data Infra, Machine Learning

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, Bash, SQL

Infra & DevOps: Terraform, Docker, Kubernetes, AWS (EC2/S3/EKS/Lambda), GCP, CI/CD, GitHub Actions, Linux

Observability: OpenTelemetry, Prometheus, Grafana, Sentry, Kafka, Distributed Tracing, Alerting

Stack: Node.js, FastAPI, PostgreSQL, Redis, MongoDB, Elasticsearch, Temporal.io, Spark, Hadoop

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)